Bayesian Causal Data Science

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MSc Social Anthropology. PhD Computer Science. **Bayesian Causal Inference**.

- Lead statistician on projects for international organizations and multinational corporations
- Data interpretation through causal models grounded in domain expert knowledge
- Researcher with publications in high-impact peer-reviewed journals
- Founder and coordinator of the Latin American Bayesian statistics community
- Optimal quantification of hypothesis uncertainty from all available evidence
- Interdisciplinary adaptability and clear communication for diverse audiences

Statistical professional in international organizations and corporations.

2025 – actual Causal Data Scientist @ *Latin America*

○ **Leading multinational beverage company.** Analysis of heterogeneous causal effects in the loyalty promotion system. Problem: promotions were designed based on fixed rules. Result: development of the strategy adopted by the team, based on a causal Bayesian network.
○ **Leading Latin American automotive marketplace (Unicorn)**. Impact evaluation of marketing efforts. Problem: A/B Testing with low statistical significance and fixed-rule attribution. Result: Development of a data-efficient methodology for the analysis of causal experiments and Data-Driven attribution.
○ **Paid Media partner for Latin American leaders.** Budget allocation across advertising channels. Problem: Poor predictive performance for high-quality conversions (low frequency). Result: Development of a hierarchical Bayesian model that significantly improved impact predictions.

2023 – 2024 Responsible for the evaluation of diagnostic tests @ *Latin America and Europe*

○ **Foundation for Innovative New Diagnostics (FIND)**. Evaluation of rapid diagnostic tests for Chagas in South America. Problem: the current methodology, expensive and slow, is a bottleneck. Result: we showed that rapid tests allow the development of a more efficient and safe methodology.
○ **European Cooperation in Science and Technology**. Consortium Action CA18208 "Novel tools for test evaluation and disease prevalence estimation". Problem: I reported to the consortium an error they were making in the evaluation of diagnostic tests. Result: publication in *PLoS Neglected Tropical Diseases*.
○ **National Health Administration (Malbrán)**. Impact experiment of Chagas strains on diagnostics. Problem: low statistical significance due to sparse data. Result: high significance achieved through a causal Bayesian model; publication in the *Journal of Molecular Diagnostics*.

2022 – actual Promoter of the continental Bayesian statistics community @ *Latin America*

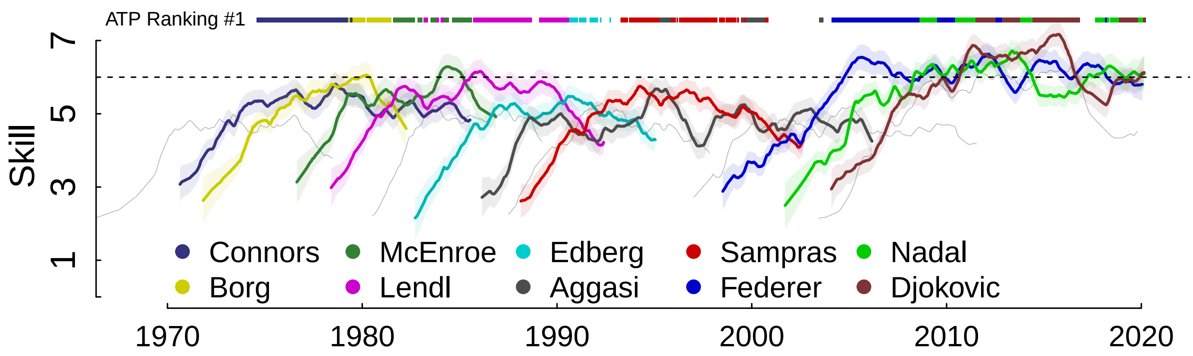
○ **Plurinational Bayes.** Problem: the adoption of Bayesian inference is lagging at the continental level. Result: continuous organization of scientific events and human resources training in Santiago 2023, Salta 2024, Bogotá 2025, Ciudad del Este 2026.
○ **Professional training in Bayesian causal inference**.
Separating correlation from causality. Optimal decisions over time.

Plurinational Bayes

Data-driven social analysis and measurement of uncertainty

2021 – 2022 The state-of-the-art skill estimator @ *Journal of Statistical Software*

Efficiently inference in causal networks with millions of nodes and irregular structures through distributed message-passing algorithms and analytical approximation methods.



2019 – 2020 Impact of the social network on skills @ *Journal of Computational Social Sicence*

Impact study of social structure on individuals' skill. Problem: classical socio-cultural studies could not evaluate the impact of social structure on human knowledge. Result: massive data from the OGS (Online Go Platform) community, graph analysis (popularity, closeness, betweenness), and the state-of-the-art skill estimator allowed for the evaluation of this impact.

2017 – 2018 Impact of team formation on skill @ *PLoS ONE*

We analyzed a community of the game Risk, between 2006 and 2013, where players could play in teams. We showed that team play leads to higher skill levels in the long term, and that playing with a stable partner accelerates skill acquisition.

Professional in social anthropology serving the public sector

2015 – 2016 Social Analysis Program of Latin American Citizenship @ *PASCAL*

Public entity created by the Media Law in Argentina for television audience measurement (ratings). My role within PASCAL was twofold: coordinator between the technical team and the social team and administrator of the database and the automated telephone survey system.

2014 Training for admission to the PhD in Computer Science @ *Universidad de Buenos Aires*

2012 – 2013 "Argentina Trabaja" Program @ Ministry of Social Development, Argentina

Assistant of the former president of the National Institute of Industrial Technology 2002-2011. My work was focused on the worker cooperatives of "Argentina Trabaja". My tasks were directed by Enrique Palmeiro, assistant of Francisco, who was appointed Pope while working with him. We conducted evaluations of the cooperatives to promote their production.

Society and culture viewed as complex systems

2011 – 2012 Prediction of parliamentary alliances @ *Université de Lausanne*

Course at the Institute of Applied Mathematics of the Faculty of Social and Political Sciences at the University of Lausanne on complexity and agent-based models, taught by Jorge Peña. Master's thesis implementing Robert Axelrod's alliance prediction model applied to European parliamentary democracies.

2010 Education in exact and natural sciences @ *Universidad de Buenos Aires*

2008 – 2009 Anthropology, complexity, simulation, and ethnography @ *Grupo Antropocaos*

Research assistant in the UBACyT project "Simulation models and anthropology, their relationship with ethnographic work". Teaching assistant in the seminar "Artificial societies and ethnography".

Education Buenos Aires University

2016 – 2022 PhD in **Computer Science.**

2012 – 2014 Master (+ BSc) in Computer Science. (Suspended after promotion to PhD)

2005 – 2009 Master (+ BSc) in **Social Anthropology**.

Teaching

2024 – actual *Bayesian Causal Inference* @ *Laboratorios de Métodos Bayesianos - ECyT UNSAM*

(1) Specification and evaluation of alternative causal models. (2) Exact and approximate inference (message-passing, variational, MCMC, SBI). (3) Inference flow, backdoor criterion, and Do-Calculus. (4) Causal ML algorithms and frameworks. (5) Temporal algorithms and generative temporal models. (6) Counterfactual inference and direct causal effects. (7) Decision-making, optimization as inference, and ergodic utility theory. (8) Socioeconomic interventions and productive financial instruments.

2019 – actual *One-on-one mentoring*. Director of 7 master's thesis in Computer Science and Data Science.

2016 – 2024 Previous courses at UBA (Depts of Computer Science and Anthropology): *Bayesian Inference*, *Algorithms & Data Structures* (I & II), *Introduction to CS*, *Functional Programming* (`Haskell`), *Computational Social Science*.

Scientific research

Articles

○ **Landfried** G., Mocskos E. *TrueSkill Through Time: reliable initial skill estimates and historical comparability in Julia, Python and R*. **Journal of Statistical Software. 2025.** 10.18637/jss.v112.i06

○ **Landfried**, G., Cairo G., Mocskos E. *Network Position and Learning Dynamics: Unveiling the Impact of Social Structure on Skill Acquisition in Online Gaming Platforms*. **Journal of Computational Social Sicence. 2025** 10.1007/s42001-025-00370-2.

○ Longhi, SA; Muñoz-Calderón, A; García-Casares, L; Irazu, L; Rodríguez, MA; **Landfried**, G; Alonso-Padilla, J; Schijman, AG; and Chagas-group *Inter-Laboratory Harmonization Study and Prospective Evaluation of the PURE-T. cruzi-LAMP Assay for Detecting Parasite Presence in Newborn Dried Blood Spots*. **The Journal of Molecular Diagnostics. 2024.** 10.1016/j.jmoldx.2024.08.007

○ Denwood, M; Nielsen, S; Olsen, A; Jones, H; Coffeng, L; **Landfried**, G; Nielsen, M; Levecke, B; Thamsborg, SM; Eusebi, P; Meletis, E; Kostoulas, P; Hartnack, S; Erkosar, B; Toft, N. *All that glitters is not gold: an interpretive framework for diagnostic test evaluation using Ascaris lumbricoides as a conceptual example*. **Plos Neglected Tropical Disease. 2024**.10.1371/journal.pntd.0012481

○ **Landfried**, G; Fernandez Slezak, D; Mocskos, E: *Faithfulness-boost effect: Loyal teammate selection correlates with skill acquisition improvement in online games.* **PLoS one. 2019.** 10.1371/journal.pone.0211014

Skills

If there is balance, it weighs less.

Software tools

`Delta Lakehouse` (`DataBricks`, `Spark`, `Streaming`, `Pipeline`, `MLflow`, `DBT` ...), `Agentic AI` (`Claude Code`, `MCP`, `Hooks`, `Subagents`, `Skills`, ...), `Python` (`Pytorch`, `QGIS`, `Pyro`, `matplotlib/Seaborn/Plotly`, `Scipy`, `Sklearn`, `Pandas`, `Numpy`, ...), `R` (`Stan`, `Tidyr`, `Tableau`, `ggplot2` ...), `Julia` (`Turing`, ...), `C++` (`MPI`), `C#` (`Infer.NET`, ...), `Java`, `Haskell`, `Bash` (`screen`, `ssh`, `vi`, `rsync`, `awk`, ...), `SQL`, `NoSQL`, `Git`, `Docker`, `Latex` (`Tikz`), `Html`, ...